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Calderon

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(54) **AUTOMATIC STEERING SYSTEMS AND METHODS**

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(2013.01)

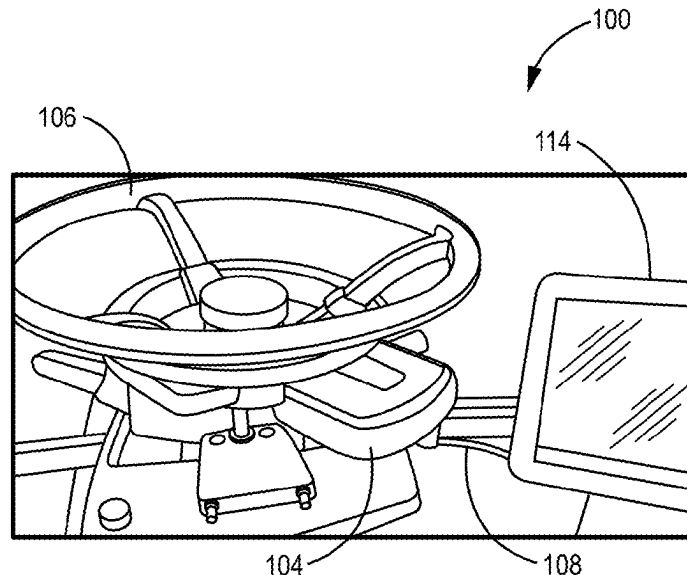
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(57) **ABSTRACT**

An automatic steering system for an agricultural vehicle includes a controller operably coupled to a steering sensor configured to provide a feedback signal that includes vehicle steering information for automatically controlling the steering device. The controller is configured to receive the feedback signal and current steering information from the steering sensor and determine desired steering rate information based on the current steering information and a desired steering position. The controller is also configured to compare a current steering rate to the desired steering rate information and disengage automatic steering control based on comparing the current steering rate to the desired steering rate information. Automatic steering systems including assisted steering devices and methods for steering an agricultural vehicle are also provided.

20 Claims, 12 Drawing Sheets



(58) **Field of Classification Search**

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See application file for complete search history.

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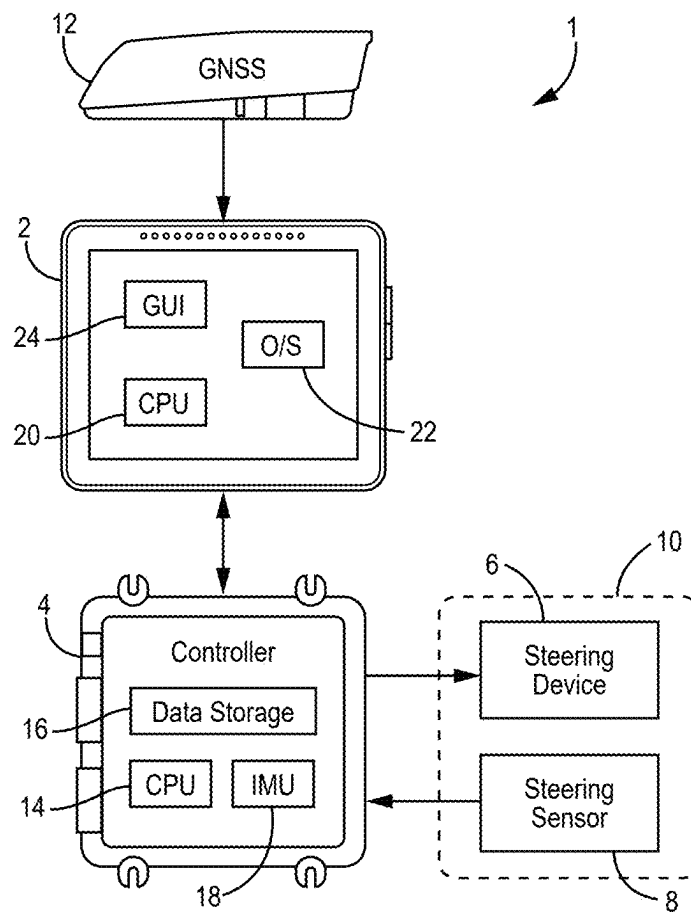


FIG. 1

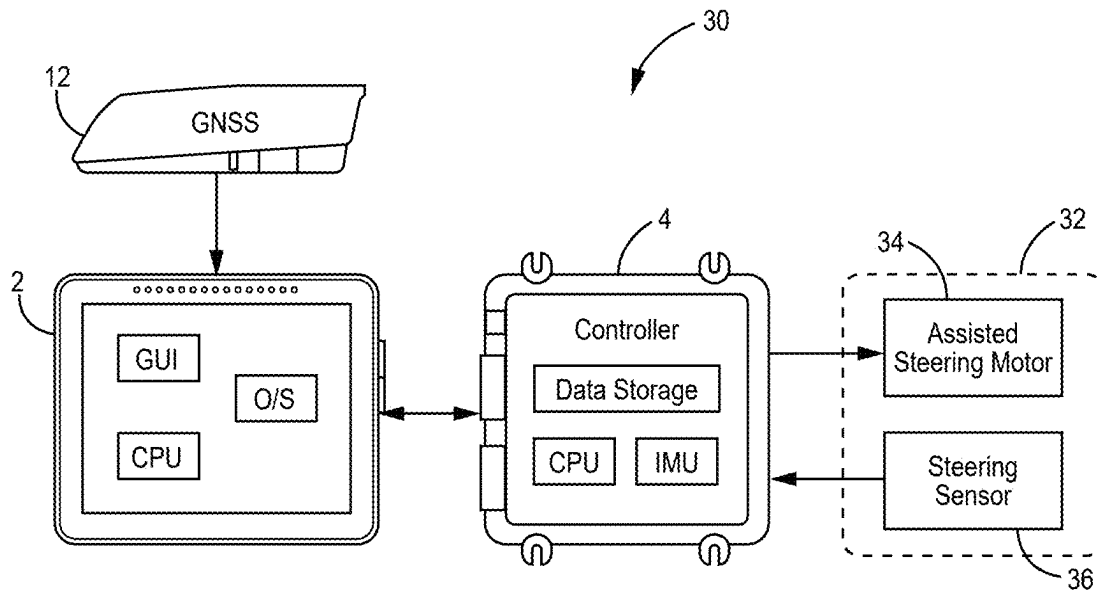


FIG. 2

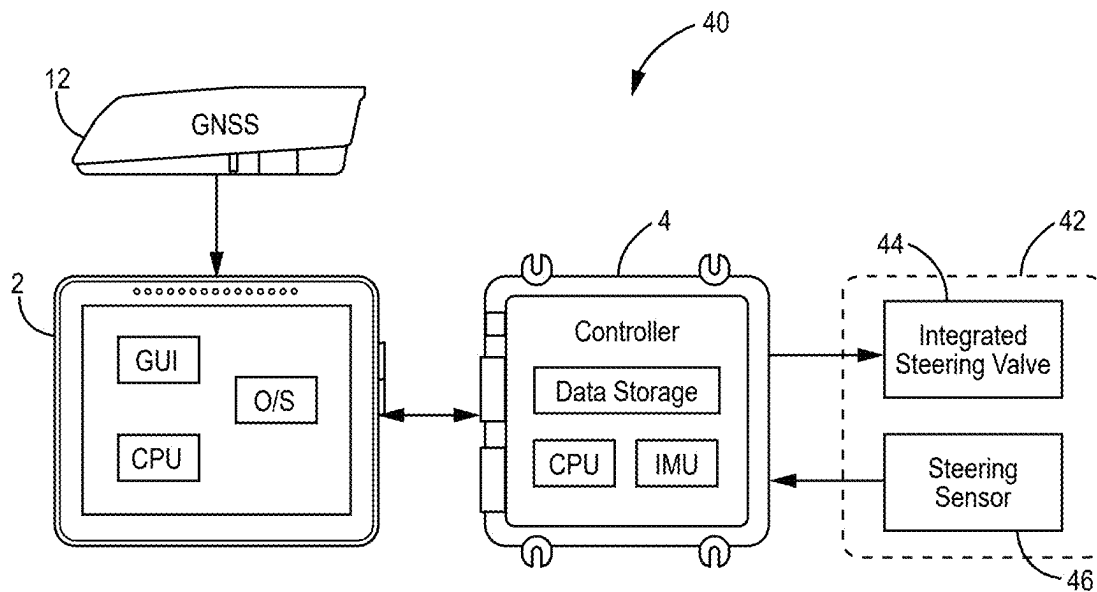


FIG. 3

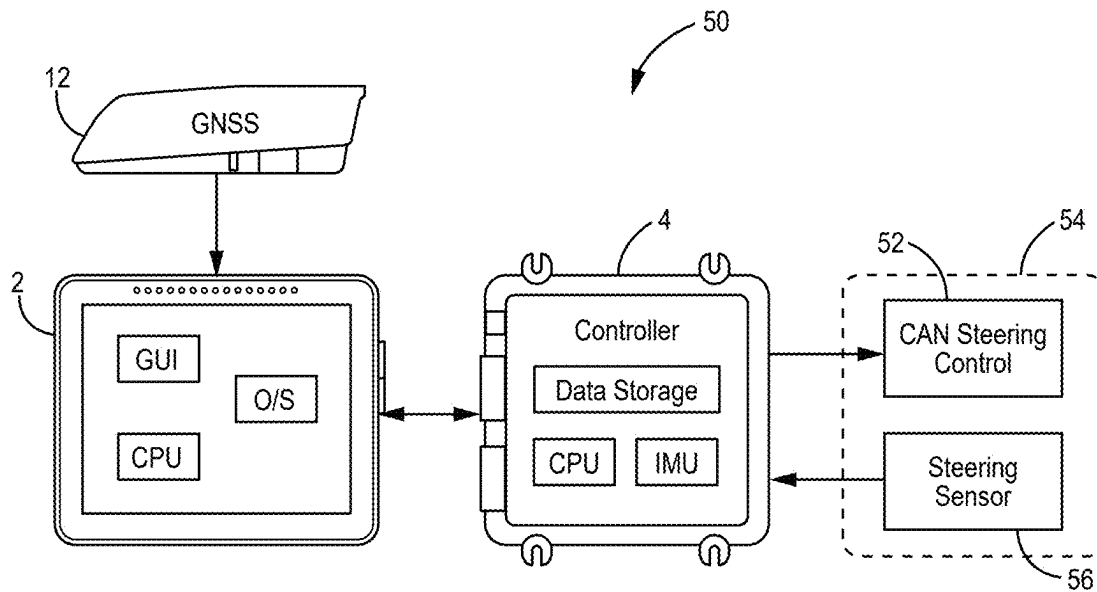


FIG. 4

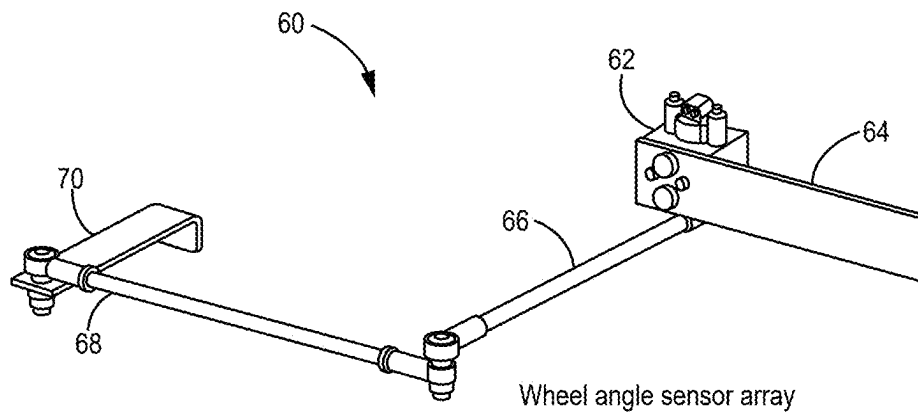
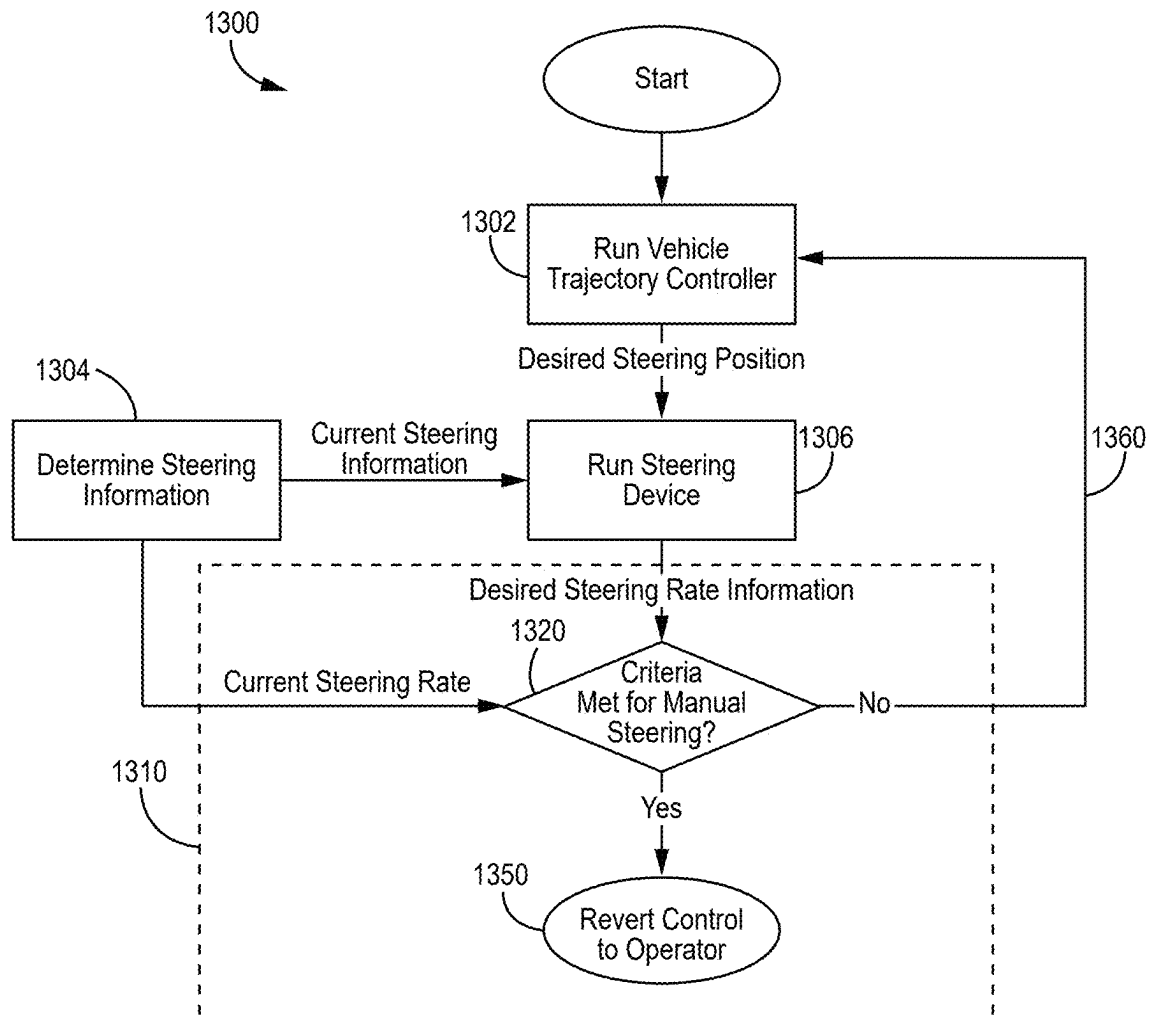
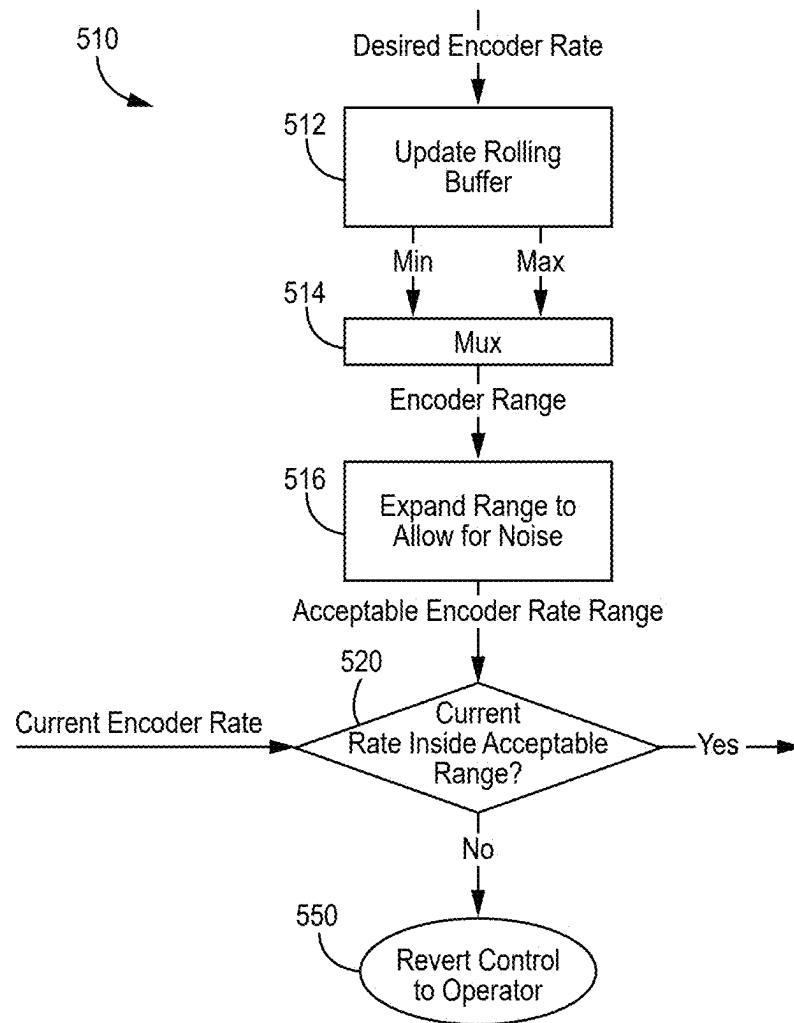


FIG. 5

**FIG. 6**

**FIG. 7**

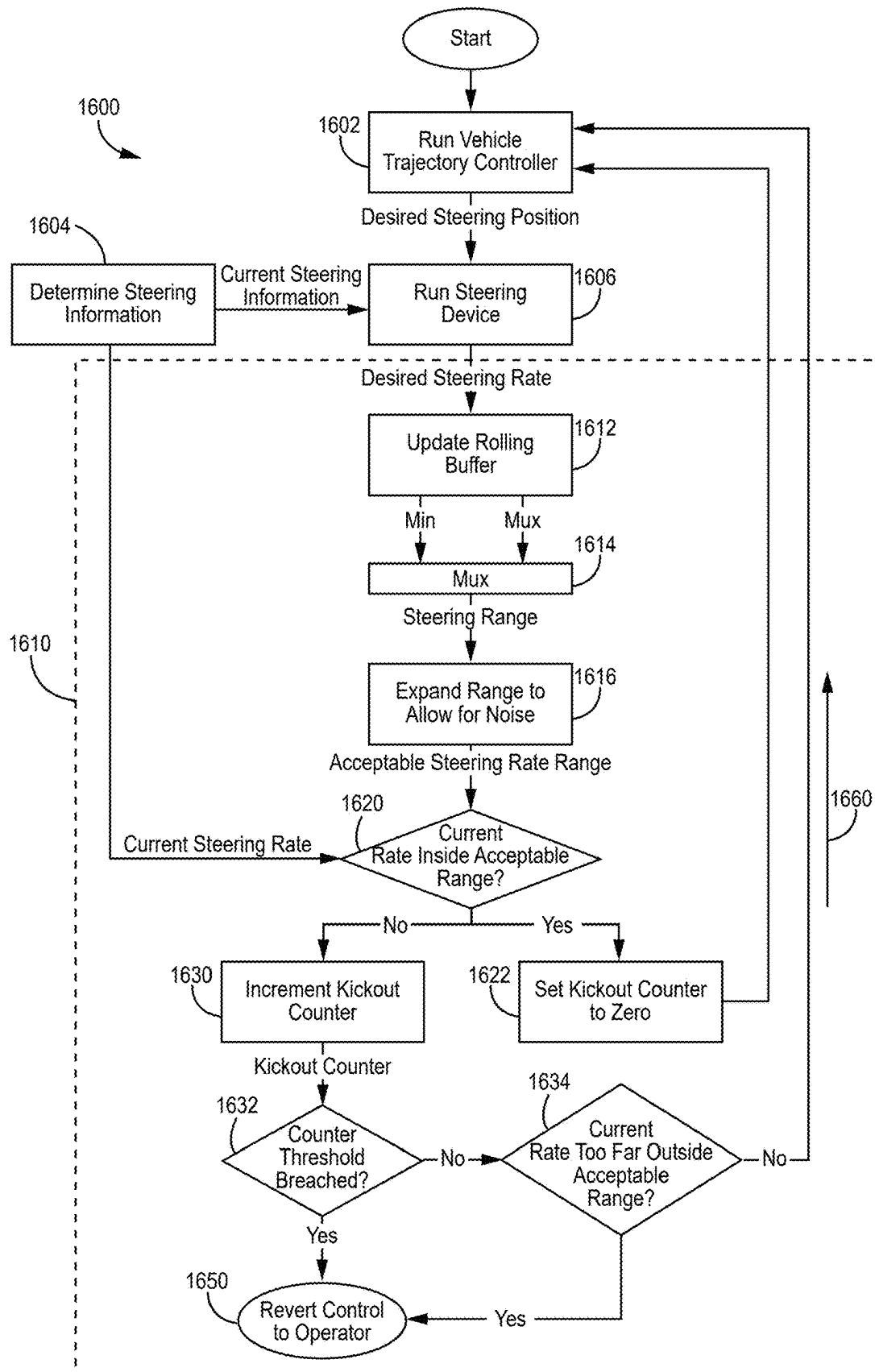


FIG. 8

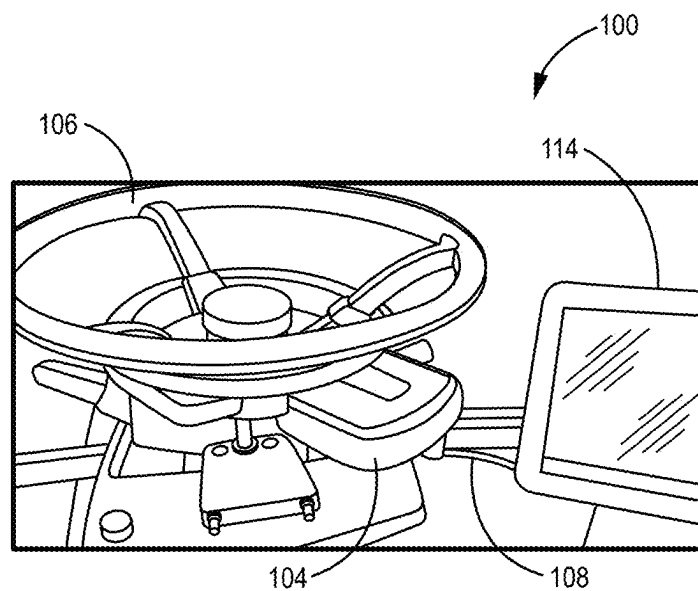


FIG. 9

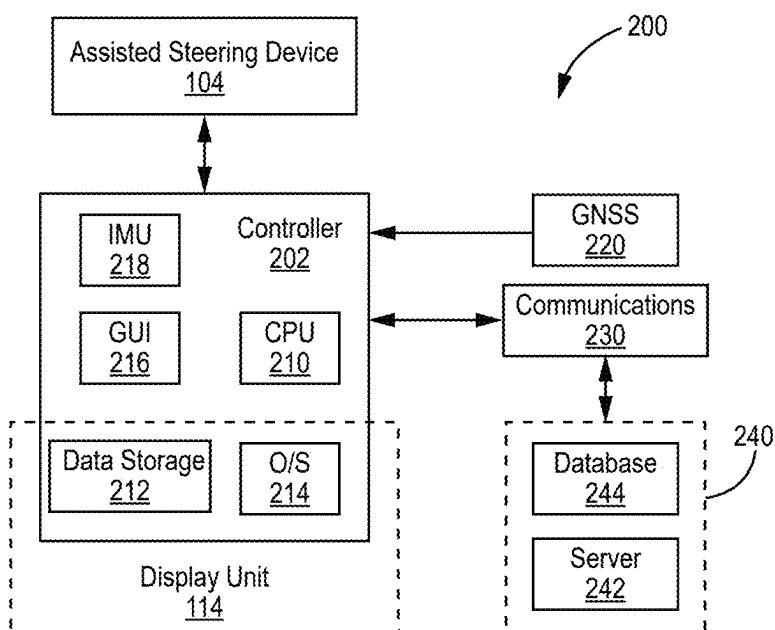
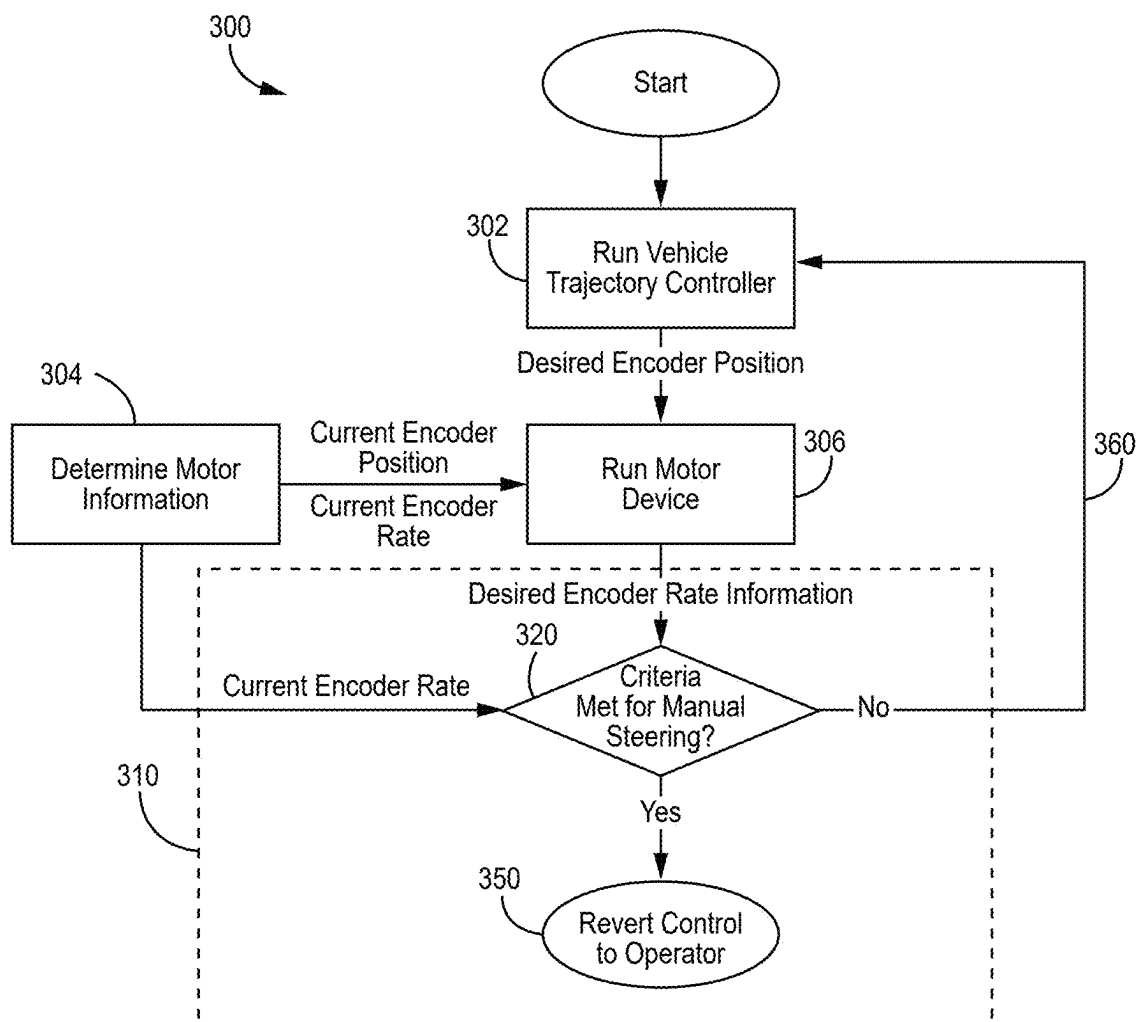
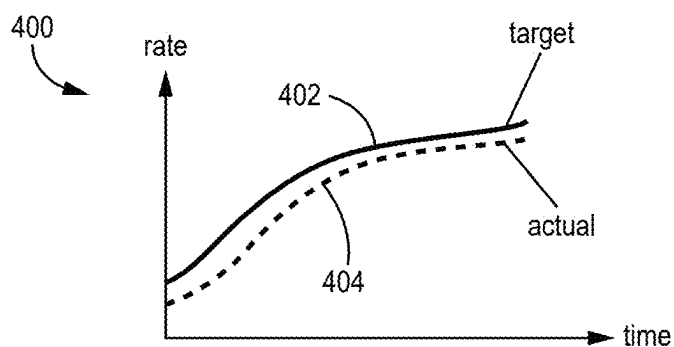
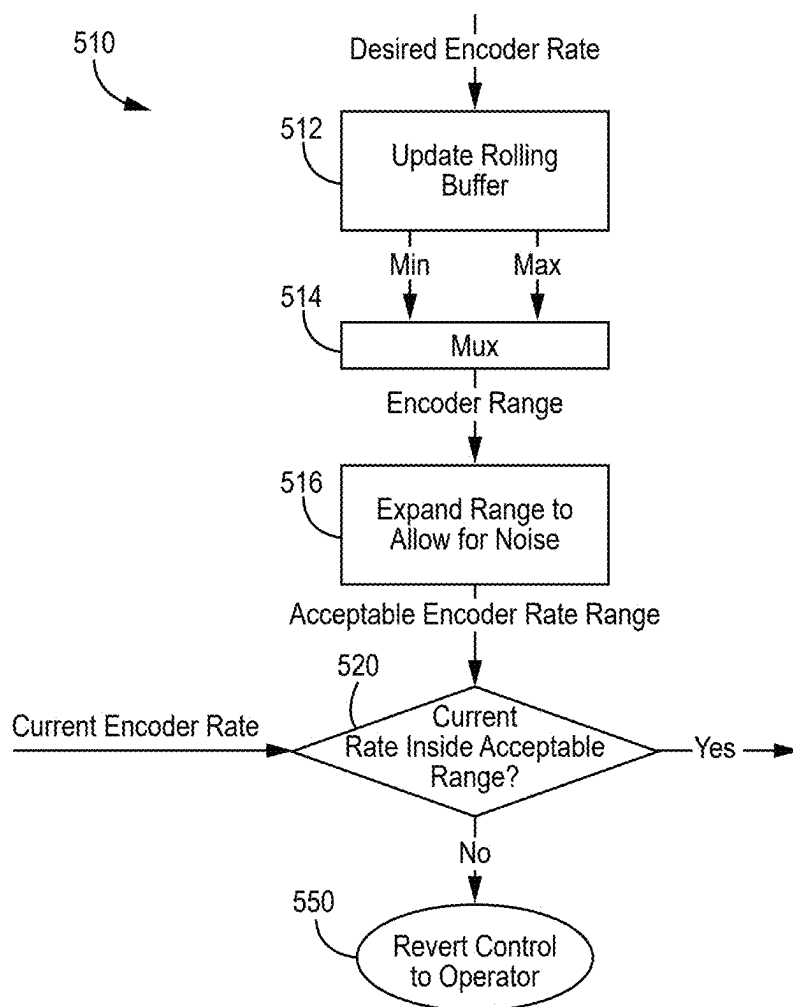


FIG. 10



**FIG. 12****FIG. 13**

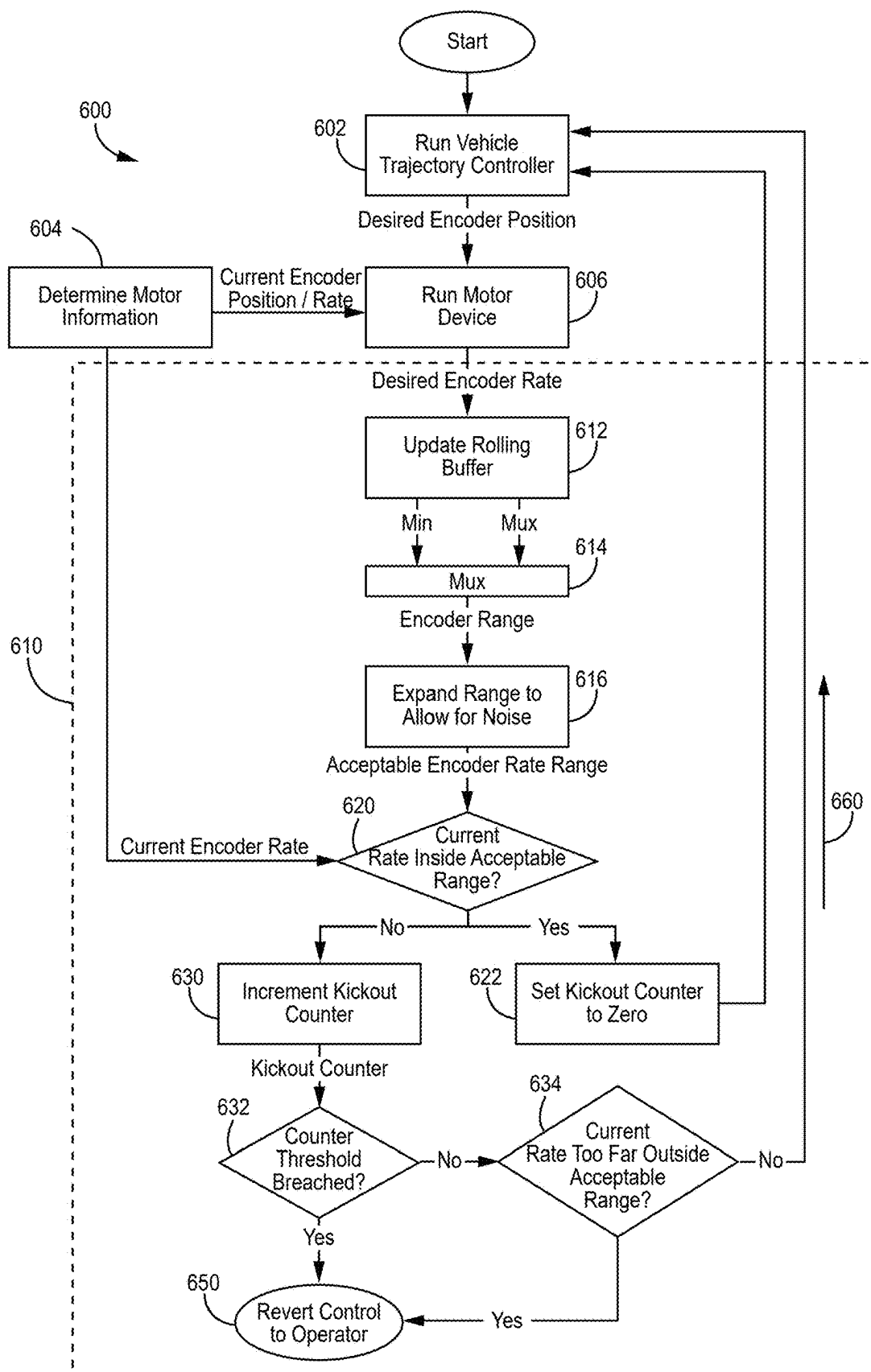


FIG. 14

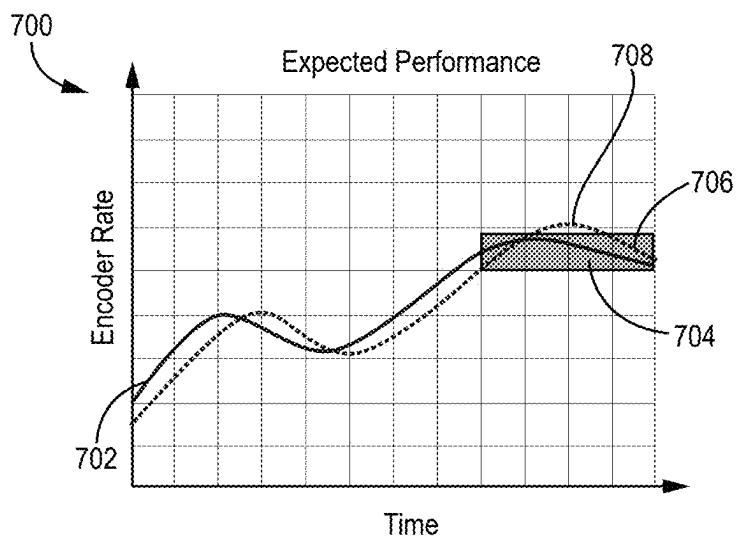


FIG. 15

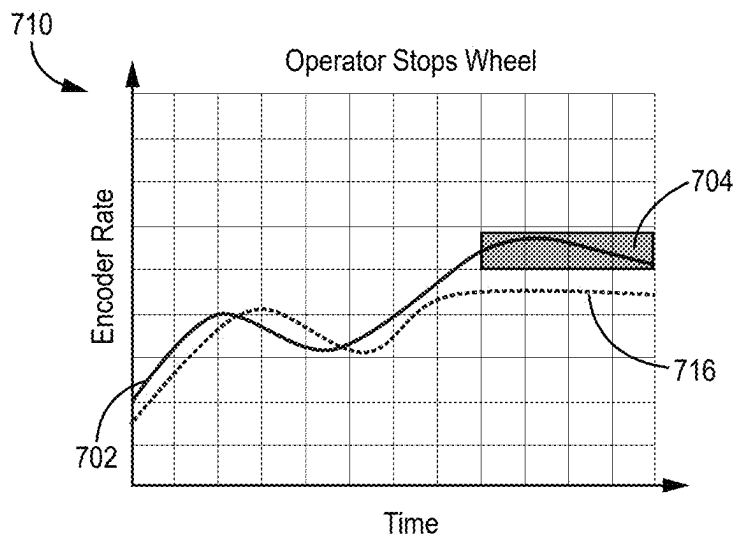
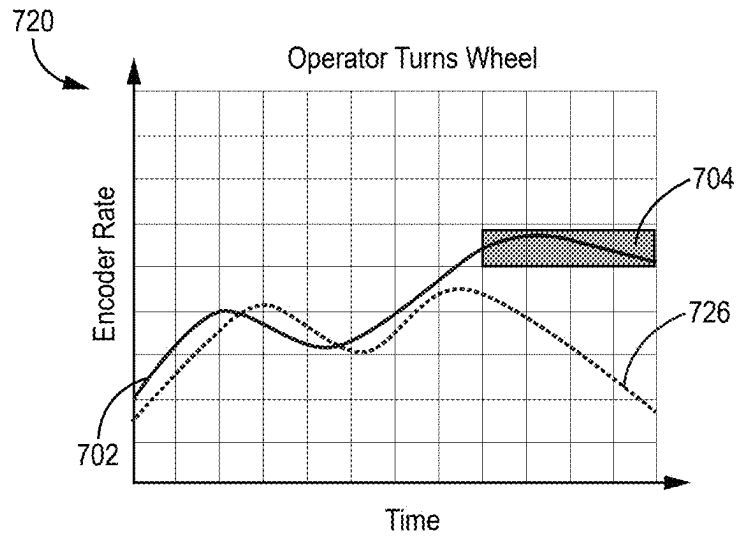
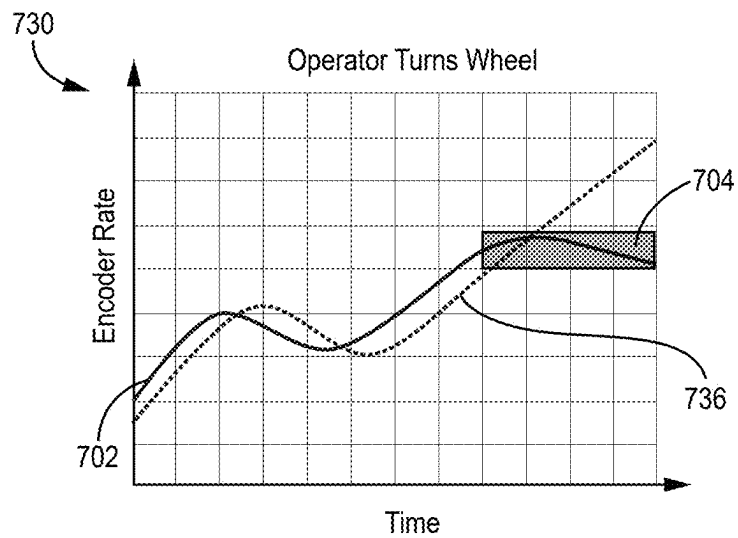


FIG. 16

**FIG. 17****FIG. 18**

AUTOMATIC STEERING SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims priority to U.S. Provisional Application No. 63/176,408 filed Apr. 19, 2021, and entitled "Automatic Steering Systems And Methods," which is hereby incorporated by reference in its entirety under 35 U.S.C. § 119(e).

BACKGROUND

The disclosure generally relates to devices, systems, and methods for improving automatic steering systems and more particularly relates to disengaging the automatic steering of agricultural vehicles.

BRIEF SUMMARY

Various devices, systems and methods related to automatic steering systems for agricultural vehicles are described herein.

An automatic steering system for a vehicle is provided according to one aspect of the disclosure. The steering system includes an assisted steering device and a controller. The assisted steering device is configured to mount to a steering wheel of a vehicle and includes a motor and an encoder. The assisted steering device is further configured to determine current encoder information. The controller is configured to operate the assisted steering device. The operation includes receiving the current encoder information from the assisted steering device, determining a desired encoder rate based on the current encoder information and a desired encoder position, and operating the motor using the desired encoder rate. The operation also includes comparing a current encoder rate to desired encoder rate information and disengaging automatic steering control when a result of the comparison meets a manual steering condition.

In various implementations of the steering system the desired encoder rate information includes an acceptable encoder rate range. In some cases the acceptable encoder rate range is based on a history of the desired encoder rate. In some cases the manual steering condition includes the current encoder rate being outside the acceptable encoder rate range. In some cases the manual steering condition further includes the current encoder rate being outside the acceptable encoder rate range for a number of consecutive comparisons exceeding a counter threshold. In some cases the manual steering condition further includes the current encoder rate being outside the acceptable encoder rate range by a difference exceeding a difference threshold. In various implementations the current encoder information includes a current encoder position. In various implementations the current encoder information includes the current encoder rate.

According to another aspect of the disclosed technology, an automatic steering system for an agricultural vehicle includes a steering sensor configured to mount to an agricultural vehicle with a steering device. The steering sensor is configured to provide a feedback signal that includes vehicle steering information for automatically controlling the steering device. The automatic steering system also includes a controller operably coupled to the steering sensor. The controller is configured to receive the feedback signal and current steering information from the steering sensor

and determine desired steering rate information based on the current steering information and a desired steering position. The controller is also configured to compare a current steering rate to the desired steering rate information and disengage automatic steering control based on comparing the current steering rate to the desired steering rate information.

In various implementations the controller is further configured to update a counter based on comparing the current steering rate to the desired steering rate information and disengage the automatic steering control when the counter exceeds a counter threshold. In various cases the controller is further configured to determine a difference between the current steering rate and the desired steering rate information and disengage the automatic steering control when the difference exceeds a difference threshold. In various implementations the desired steering rate information comprises a desired steering rate for operating the steering device. In various implementations the desired steering rate information comprises an acceptable steering rate range. In various implementations the controller is further configured to determine the acceptable steering rate range based on a history of a desired steering rate for operating the steering device.

In various implementations the automatic steering system also includes an assisted steering device configured to mount to a steering wheel of a vehicle. The assisted steering device includes a motor and an encoder. In addition, the controller is operably coupled to the assisted steering device, the steering sensor includes the encoder, the current steering information includes current encoder information, the desired steering rate information includes desired encoder rate information, and the current steering rate includes a current encoder rate.

A method for steering an agricultural vehicle is provided according to another aspect of the disclosed technology. The method includes steering the agricultural vehicle with an automated steering device mounted to the vehicle and receiving a feedback signal that includes current steering information with a controller configured to operate the automated steering device. The method also includes comparing the current steering information to desired steering information and disengaging automatic steering control when a result of the comparison meets a manual steering condition.

According to various implementations, steering the agricultural vehicle includes steering with an assisted steering device mounted to a steering wheel of the vehicle. In some cases the assisted steering device includes the automated steering device. The method can further include receiving the feedback signal from the assisted steering device. In various cases the assisted steering device includes a motor and an encoder, the current steering information includes a current encoder rate, and the desired steering information includes an acceptable encoder rate range.

In some cases the manual steering condition includes the current encoder rate being outside the acceptable encoder rate range. In various implementations the manual steering condition further includes the current encoder rate being outside the acceptable encoder rate range for a number of consecutive comparisons exceeding a counter threshold. In various implementations the manual steering condition further includes the current encoder rate being outside the acceptable encoder rate range by a difference exceeding a difference threshold.

According to the various implementations, automatic steering systems include one or more controllers and/or computers that are configured to perform particular opera-

tions or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the operations or actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions. Methods for steering an agricultural vehicle can be carried out with the use of one or more such controllers, computers, and/or computer programs.

Other examples of the disclosed implementations include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods. While multiple embodiments are disclosed, still other embodiments of the disclosure will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the disclosed apparatus, systems and methods. As will be realized, the disclosed apparatus, systems and methods are capable of modifications in various obvious aspects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an automatic steering system for an agricultural vehicle according to an implementation.

FIGS. 2-4 are schematic views of automatic steering systems for an agricultural vehicle according to various implementations.

FIG. 5 is a perspective view of a wheel angle sensor kit according to an implementation.

FIG. 6 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIG. 7 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIG. 8 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIG. 9 is a perspective view of an automatic steering system mounted to a steering wheel of an agricultural vehicle according to an implementation.

FIG. 10 is a system diagram of an automatic steering system according to an implementation.

FIG. 11 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIG. 12 is a graph comparing plots of targeted and actual encoder rates according to various implementations.

FIG. 13 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIG. 14 is a flow diagram illustrating operation of an automatic steering system according to an implementation.

FIGS. 15-18 are graphs comparing plots of targeted and actual encoder rates according to various implementations.

DETAILED DESCRIPTION

The various embodiments disclosed and contemplated herein relate to automatic steering systems for agricultural vehicles. More specifically, various implementations of the disclosed embodiments enable transitioning from an automatic steering mode to a manual steering mode. Embodiments can further be described as disengaging from or overriding the automatic steering mode, reverting to manual

steering, and/or providing a kickout process to kickout an automatic steering control and give steering control back to a human operator.

According to one aspect of the disclosure, an automatic steering system disengages from automatic steering control when an operator handles a vehicle steering wheel. For example, in some cases the system detects that the operator is manually turning the steering wheel and disengages the automatic steering mode in response. As another example, in some cases the system disengages automatic steering control after detecting that the operator has simply grabbed or is holding the steering wheel without turning the wheel. Reverting to manual steering control under these circumstances can be especially useful for enabling manual control while continuing to drive the vehicle straight ahead.

According to another aspect of the disclosure, an automatic steering system considers one or more criteria (also referred to herein as conditions) to determine whether to disengage automatic steering. In some implementations, the steering system evaluates the length of time that an operator handles a steering wheel before disengaging automatic steering. As one example, in some cases a steering system transitions from automatic to manual steering after an operator has held a steering wheel for a certain period of time. In some implementations, the system compares an expected turning rate with an actual turning rate to determine whether to revert to manual steering. For example, in some cases the system determines if the vehicle (e.g., the steering wheel) is turning more quickly or slowly than expected or in a different direction than expected. In some cases the system transitions from automatic to manual steering when at least one of the actual turning speed and direction is different from the expected turning speed and direction. In such cases differences between automatic steering commands and actual movement of the steering wheel help determine if an operator is physically overriding the automatic steering, thus indicating a desire to switch to manual steering.

According to various implementations, an automatic steering system determines whether to disengage automatic steering without needing to consider the force applied to the steering wheel by the operator. Although sensing torque applied to a steering drive motor is a potentially useful way to determine an operator's desire for manual steering, solely relying on this technique also requires limiting the output torque of the drive motor so that the operator can physically override the automatic steering. As will be further discussed herein, various implementations of the disclosed technology advantageously detect operator handling of a steering wheel without per se sensing the torque and/or current draw of the steering drive motor, thus avoiding the corresponding need to limit the output torque of the steering drive motor.

According to another aspect of the disclosure, various implementations of an automatic steering system use a feedback signal with steering information to determine whether to disengage or kickout the automatic steering control. In various implementations the steering information in the feedback signal indicates an actual turning rate of the vehicle's steering wheel and/or one or more vehicle wheels. For example, in some cases the feedback signal indicates how fast the vehicle steering wheel is turning and in which direction. In various implementations the steering information includes a steering position of the vehicle's steering wheel and/or one or more vehicle wheels.

In certain cases, the determination to disengage from automatic steering control at least partly depends on whether one or more characteristics of the feedback signal meet one or more manual steering criteria or conditions. For example,

5

in certain implementations a determination that the actual turning rate is zero indicates that the vehicle operator is holding the steering wheel and meets a manual steering condition. In various implementations determining whether manual steering condition(s) are met involves comparing steering information from the feedback signal with expected (also referred to herein as “desired” or “target” or “targeted”) steering information. As an example, in various implementations the actual turning rate indicated by the feedback signal is different from an expected or targeted rate of turning, which can indicate that an operator is physically turning the steering wheel. As another example, the steering position (e.g., of the vehicle’s wheels or steering wheel) indicated by a feedback signal may be different from an expected or desired steering position. An automated steering system can thus disengage automatic steering control under one or more of these or other circumstances.

In various implementations the feedback signal(s) are generated by one or more steering sensors. As an example, in some cases a steering encoder or another sensor that is part of an assisted steering system may generate a steering feedback signal. In some cases one or more feedback signals are received from a steering sensor through a Controller Area Network (CAN) bus or another vehicle network. For example, a vehicle steering sensor may generate and transmit a feedback signal to the CAN bus that includes a steering wheel or vehicle wheel position and/or turning rate. In some cases an automated steering system may receive one or more feedback signals from a wheel angle sensor that provides a wheel angle position or rate. In some cases a factory installed or aftermarket wheel angle sensor generates a steering feedback signal that can be used by an automatic steering system to control, for example, an integrated valve steering mechanism. As another example, in some cases a vehicle’s steering wheel may be equipped with one or more rotational sensors that generate feedback signals based on the wheel’s rotational position and/or rate of turning. Other sources for feedback about the current steering angle and/or rate of change are also possible. Turning to the drawings, FIG. 1 is a schematic view of an automatic steering system 1 for an agricultural vehicle according to various implementations. As depicted in this particular example, the automatic steering system 1 includes a display unit 2 in communication with a steering controller 4. The steering controller 4 is coupled in communication with a vehicle steering device 6 and a vehicle steering sensor 8. According to various implementations the steering device 6 and the steering sensor 8 are part of an integrated steering mechanism 10.

In various implementations, the automatic steering system 1 is also operationally integrated with a GNSS or GPS unit 12, such as a GPS 7500, such that the system 1 is configured to input positional data for use in defining boundaries, locating the vehicle, plotting guidance, and the like, as will be readily appreciated. As an example, in some cases the controller 4 is configured to determine (e.g., estimate) the current position and/or trajectory of the vehicle using data from the GNSS/GPS unit 12.

According to various implementations, the controller 4 includes one or more processing and computing components for implementing various technologies disclosed herein and elsewhere. As shown in FIG. 1, some of the components of the controller 4 can include a CPU or processor 14, data storage 16, and an optional inertial measurement unit (IMU) 18. The controller 4 is communicatively coupled with the display 2. In some cases the display 2 includes its own processing and computing component(s), such as a CPU or other processor 20, which runs an operating system 22 that

6

provides a graphical user interface (GUI) 24. One example of a display unit is the display unit sold under the name InCommand® by Ag Leader Technology of Ames, Iowa. It is appreciated that the display 2 may include a touchscreen and/or be equipped with interface components such as a keyboard, mouse, and/or other device for interacting with the system 1.

It should be appreciated that the controller 4 may be located in one of many places according to various implementations. For example, in some cases the controller is a discrete physical component mounted separately from other components. In some cases the controller is mounted within and/or is integral to another component, such as the display unit 2, the steering mechanism 10, or another component. In the depicted example the display 2 is provided separately from the controller 4. It should be appreciated that the various components of the display 2 and the controller 4 are in operational communication with one another via wired or wireless connections and are configured to perform processes and execute commands to implement operations and features described herein.

According to various implementations, the controller 4 is configured to determine a desired trajectory and control the steering device 6 to steer the vehicle along the desired trajectory. As an example, in some cases the controller is configured to determine a vehicle curvature that will guide the vehicle from its current position to a new position along the desired trajectory. In various implementations the controller determines the vehicle curvature as a desired steering angle or position of the vehicle wheel(s) and/or steering wheel. The controller is configured to then determine a turning rate (e.g., in terms of speed and direction) to bring the steering wheel or vehicle wheel(s) from its current position to the desired position within a particular amount of time. It is appreciated that in various implementations, one or more controller is provided.

In various implementations the system 1 also includes a communications component (not shown) configured for sending and receiving data to one or more cloud-based locations for processing and storage. As examples, data may be transmitted to and from a remote server, database, and/or other cloud computing components readily understood in the art. Such connections by the communications component can be made wirelessly via understood internet and/or cellular technologies such as Bluetooth, Wi-Fi, LTE, 3 G, 4 G, or 5 G connections and the like. It is understood that in certain implementations, the communications component and/or cloud components comprise encryption or other data privacy components such as hardware, software, and/or firmware security aspects. In various implementations, the operator or enterprise manager or other third parties are able to receive notifications such as adjustment prompts and confirmation screens on their mobile devices, and in certain implementations can review the plotted guidance paths and make adjustments via their mobile phones.

As depicted in the example of FIG. 1, in some cases the steering device 6 and the steering sensor 8 are part of an integrated steering mechanism 10. For example, the integrated steering mechanism 10 can in some cases be an assisted steering device that includes a motor drive unit (MDU) with a drive motor configured to rotate the vehicle’s steering wheel in response to commands from the controller. In such a case, the steering sensor 8 is provided by a motor encoder that senses rotation of the motor using, for example, a Hall-effect sensor. The MDU generates and sends a feedback signal to the controller 4 based on the sensor output.

7

According to various implementations, the steering device **6** may be a hydraulic valve that is part of an integrated hydraulic steering system. As another example, in some cases the steering device **6** may be an OEM or factory-installed steering control system (e.g., including a hydraulic, electric, electrohydraulic, or another type of steering device) that is controllable through a Controller Area Network (CAN) bus or another vehicle network. In such examples, the steering sensor **8** may be provided as part of an integrated steering system or an OEM steering control system or may be provided separately as a dealer or after-market installed sensor.

According to various implementations, the steering sensor **8** may be one of various types of sensors suitable for generating one or more feedback signals conveying steering information. As noted above, in some cases the steering sensor **8** is provided by a motor encoder. In various cases the steering sensor **8** is a factory installed or aftermarket wheel angle sensor that generates a steering feedback signal. As another example, in some cases a vehicle's steering wheel may be equipped with one or more rotational sensors that generate feedback signals based on the wheel's rotational position and/or rate of turning. In some cases the steering sensor **8** may include a strain gauge that changes resistance as it bends. Measuring the resistance enables determination of the amount of bend in the sensor, which can enable measuring an angle of a wheel relative to a vehicle when attached to a wheel linkage. Such a measurement could also provide the position of the wheel (also called the vehicle's wheel angle). In various other cases, the steering sensor **8** may be a pressure sensor. For example, the sensor may measure the changing pressure in a valve attached between the wheel and the vehicle. Turning the wheel changes the wheel angle, which in turn changes the valve pressure, which can be sensed and sent to the controller as a feedback signal. Of course other types of steering sensors are possible and implementations of the disclosed technology are not limited to any one type of steering sensor.

FIGS. 2-4 are schematic views of various automatic steering systems for an agricultural vehicle according to various implementations. The steering systems include several components in common with the system **1** shown in FIG. 1, but in each case include a differing type of automated steering device. FIG. 2 illustrates an example of an automatic steering system **30** that includes an assisted steering device **32** in communication with the steering controller **4**. The assisted steering device **32** includes a steering motor **34** and a steering sensor **36**. One example of an assisted steering device is a motorized system that attaches directly to a vehicle's steering wheel. The system's motor simulates a human physically turning the steering wheel to turn the vehicle's wheels. In some cases the steering sensor **36** is provided by a motor encoder that senses rotation of the steering motor **34** using, for example, a Hall-effect sensor. As shown in FIG. 2, the steering motor **34** and steering sensor **36** are in some cases provided as part of an integrated assisted steering device **32**. It should be appreciated that in various implementations the steering motor and steering sensor may be provided separately as well.

FIG. 3 illustrates a schematic diagram of another automatic steering system **40** that includes a steering mechanism **42** with an integrated steering valve **44** according to various implementations. As an example, in some cases the steering valve **44** is a hydraulic valve that attaches directly to the vehicle's wheels or steering linkage. The controller **4** is configured to send the valve **44** a command that actuates the valve, thus pushing on the vehicle's wheel and causing it to

8

turn. In various cases the integrated steering valve is built into an agricultural vehicle, such as a tractor and movement of the valve does not result in the steering wheel turning in the cab. In some cases an agricultural vehicle may be equipped with a steering valve at the factory, while in some cases an aftermarket valve may be installed. The steering mechanism **42** also includes a steering sensor **46**, which in some cases is housed and mounted to the vehicle separately from the steering valve **44**. In some cases the steering sensor **46** may be implemented by one of the sensors described herein, or by any other sensor of suitable design.

FIG. 4 illustrates a schematic diagram of another automatic steering system **50** that includes a steering mechanism **52** with an OEM or factory-installed steering control **54** (e.g., including a hydraulic, electric, electrohydraulic, or another type of steering device) that is controllable through a Controller Area Network (CAN) bus or another vehicle network. The steering mechanism **52** also includes a steering sensor **56**, which in some cases is housed and mounted to the vehicle separately from the steering control **54**. In some cases the steering sensor **56** may be implemented by one of the sensors described herein, or by any other sensor of suitable design.

Turning to FIG. 5, a perspective view of a wheel angle sensor kit **60** is depicted according to various implementations. The kit **60** generally includes a wheel angle sensor assembly **62** mounted to a sensor bracket **64** for coupling the sensor to a stationary point on an agricultural vehicle, such as a fixed point on the chassis or axle housing. A linkage arm **66** and a linkage rod **68** couple to the sensor assembly **62** to a linkage rod bracket **70**, which can be mounted to a moving part of the steering system, such as a wheel support or steering linkage. As the vehicle's wheel turns, the interconnected linkages actuate the sensor assembly **62**, thus generating a feedback signal corresponding to the wheel's steering position and/or turning rate.

FIG. 6 is a flow diagram illustrating operation **1300** of an automatic steering system according to various implementations. Among other things, the automatic steering system's operation includes running **1302** a vehicle trajectory controller to generate a desired steering position for adjusting the steering of a vehicle. The desired steering position may correspond to, for example, a wheel position or a steering wheel position that will result in a particular steering curvature. As shown in FIG. 6, the steering system also determines **1304** steering information. In some cases, a steering sensor, such as a motor encoder, a wheel angle sensor, a pressure sensor, strain gauge sensor, or another type of sensor measures various properties of a vehicle's steering wheel, wheels, and/or steering devices and reports the information. In various implementations the determined steering information includes current steering information. Examples of current steering information are current encoder information for a motor encoder, current resistance measurements for a strain gauge sensor, current readings for a potentiometer or Hall effect sensor, and the like. In some cases the current steering information includes a current steering position, a current steering rate, or both. The desired steering position is then used along with one or both of the current steering rate and the current steering position to run **1306** a steering device to change the vehicle's steering trajectory. As an example, in some cases a desired steering position is used along with one or both of a current steering position and a current steering rate to determine a desired steering rate for moving the steering device (and connected steering linkages, steering wheel, wheels, etc.) to the desired position.

In the depicted implementation, the operation of the automatic steering system includes a kickout analysis or process **1310** that determines whether to disengage automatic steering control of the vehicle. In general, the kickout process **1310** includes analyzing **1320** a current steering rate and desired steering rate information. In various implementations, the desired steering rate information is based on the desired steering position and one or both of the current steering position and current steering rate. As an example, in some cases the desired steering rate information includes a desired steering rate generated to run **1306** the steering device as previously discussed. In some cases the desired steering rate information may be based on additional processing of a desired steering rate and may include, for example, a history of desired steering rates.

When the analysis determines that condition(s) for manual steering control are met, the automatic steering is disengaged and control reverts **1350** to the operator. In some circumstances the analysis **1320** may instead determine that manual steering criteria are not met. In these cases the operational flow loops back **1360** to continue automatic steering control by running **1302** the vehicle trajectory controller until a subsequent kickout routine **1310** begins. It will be appreciated that in various implementations the analytical logic may be switched while retaining the same substantive analysis. For example, in some cases the analysis may instead ask if condition(s) for automatic steering control are met, with a true answer leading to the continuation of automatic steering control and a false answer leading to manual steering control.

According to various implementations, the automatic steering operation **1300** is carried out by an automatic steering system such as one of the systems shown in FIGS. **1-4**. In such cases the steering system controller(s) operate(s) as one or both of the steering device controller and trajectory controller in FIG. **6**. In various implementations, one or more controllers are provided to accomplish the functionality described herein, as would be appreciated.

In some cases the steering system controller also determines **1320** if manual steering conditions or criteria are met and reverts control **1350** to the operator or loops control back **1360** to the vehicle trajectory controller process **1302**. According to various implementations, one or more additional controllers may carry out portions of the operation **1300**. As just one example, in some cases an entirely separate vehicle trajectory controller determines the vehicle's position and generates a desired steering position for correcting the vehicle's trajectory.

Various criteria, conditions, and/or analyses are used to evaluate the current steering rate and the desired steering rate information (e.g., desired steering rate) in various implementations. As an example, in some cases the steering system evaluates the desired steering rate information and the current steering rate to determine how the vehicle's steering is tracking with the trajectory determined by the vehicle trajectory controller. In some cases differences between the desired and current steering rates indicate that an operator is holding the steering wheel and/or overriding the automatic steering by physically turning the steering wheel. In such cases the system disengages the automatic steering control and reverts control **1350** to the operator.

In some cases a lag exists between determining a desired steering position and desired steering rate and the steering system responding accordingly. Various implementations of an automatic steering system address such a lag by comparing a current steering rate with desired steering rate information that includes a history of desired steering rates.

Such an analysis enables the steering system to determine if the steering rates reported from the steering sensor are tracking in the expected direction.

FIG. **7** depicts one example of comparing the current steering rate with desired steering rate information that includes a history of desired steering rates according to various implementations. FIG. **7** depicts part of a kickout process **1510** that generates a desired or acceptable range of target steering rates and then determines **1520** whether the current steering rate is within the range of desired steering rates. A determination that the current steering rate is not inside (i.e., outside) the acceptable range is an indication that manual steering control may be desired and the system reverts steering control **1550** to the operator. In contrast, a determination that the current steering rate is within the acceptable range indicates that automatic steering should continue.

An acceptable range of desired or targeted steering rates can be generated in various suitable manners according to the implementations disclosed and contemplated by this disclosure. In the implementation depicted in FIG. **7**, the automatic steering system updates **1512** a rolling buffer with new desired steering rates as they are received. In various cases the buffer contains the last twenty desired steering rates. The buffer can collect a larger or smaller number of rates in various implementations. In some cases an expected range of steering rates is then determined **1514** using the maximum and minimum rates in the buffer.

In some implementations the range of steering rates is expanded **1516** by some amount to provide tolerance for system noise. As an example, the minimum rate of the range may be lowered by some amount and/or the maximum rate of the range may be increased by some amount. According to various implementations, the maximum rate is increased by 10% of its value and the minimum rate is lowered by 10% of its value. Different amounts of tolerance can be used depending upon the requirements or preferences of a particular implementation. Further implementations are of course possible, regardless of limitations on controller rate, length of the rolling buffer, minimum tolerance or otherwise.

Once the steering system has generated an expected range of acceptable steering rates, the system determines **1520** whether the current steering rate is within the acceptable range and either maintains automatic control or reverts to manual control as noted above.

FIG. **8** is a flow diagram illustrating operation **1600** of another automatic steering system according to various implementations. As with other examples described herein, the automatic steering system includes running **1602** a vehicle trajectory controller to generate a desired steering position for adjusting the steering of the vehicle. The steering system also determines **1604** steering information (e.g., with a steering sensor) to determine the current steering position. The desired steering position and one or both of the current steering position and current steering rate are then used to run **1606** a steering device to change the vehicle's steering trajectory. In various implementations one or both of the current steering position and current steering rate and the desired steering position are used to determine a desired steering rate for adjusting the vehicle's steering. In some cases running the vehicle trajectory controller, determining steering information, and running the steering device are carried out in the same manner as the system operation **1300** described with respect to FIG. **6**.

The steering system operation also includes a kickout process or procedure **1610** that determines whether to give manual control **1650** to an operator or to loop back **1660** to

11

continue automatic steering control by running **1602** the vehicle trajectory controller. As with the example of FIG. 7, the kickout process **1610** includes generating an acceptable range of target steering rates and then determining **1620** whether the current steering rate is within the acceptable range. To generate the acceptable range of steering rates, the automatic steering system determines **1614** the maximum and minimum rates from a rolling buffer **1612** of desired steering rates and then expands **1616** the range by some amount to provide tolerance for system noise.

As discussed elsewhere, determining that the current steering rate is outside the desired steering range may indicate that control should revert to manual steering. In some cases, it may be desirable delay the switch to manual control in order to, for example, confirm the determination to disengage automatic control. As an example, in some implementations steering control is switched to a manual mode only after detecting multiple subsequent instances of the current steering rate being outside the acceptable steering range. The example kickout process **1610** in FIG. 8 includes incrementing **1630** a kickout counter each time the current steering rate is outside the acceptable range of steering rates. Upon exceeding a counter threshold **1632**, the steering system reverts control **1650** to the operator.

According to various implementations, the counter threshold **1632** can be set by the operator running the system. In some cases setting the counter threshold to a low value allows for easily and quickly switching to manual steering control. In some implementations a higher value for the counter threshold results in a longer kickout time but may result in fewer false kickouts. In some cases false kickouts occur when the steering wheel is bumped or an operator rests their hand on the wheel. In some cases field conditions bump the vehicle wheels. The movement can at times feed back into the steering column and be falsely interpreted by the system as the operator moving the steering wheel.

In some cases the counter threshold **1632** can be fixed to require any number of kickouts suitable for a particular operating environment. In implementations such as those discussed above, an operator can set the counter threshold **1632** to a value within a predefined range. In some cases the counter threshold can be any suitable value from a single instance to 50 or 100 or more instances. It should be appreciated as well that the desirable number for the threshold may depend upon the operating frequency of the system. In some cases the threshold can be set to a number from 3 to 43, which corresponds to a time range of 0.06 seconds up to 0.8 seconds when operating at 50 Hz.

As already noted, in some cases implementations of an automatic steering system will maintain automatic steering control, loop back to running the vehicle trajectory controller, and then conduct a new kickout analysis with updated steering rates. For example, FIG. 8 illustrates how the steering system operation loops back **1660** to running the vehicle trajectory controller **1602** in certain circumstances. The kickout process **1610** illustrates two situations in which automatic control is maintained in this way. In some cases the kickout process may determine **1620** that the current steering rate is actually within the acceptable range of desired steering rates. Under these circumstances, the automatic steering system sets the kickout counter to zero **1622** and operational control flows back **1660** to running the vehicle trajectory controller **1602**.

As another example, in some implementations a kickout process **1610** further considers the degree to which a current steering rate is outside the acceptable range of rates. In the

12

example shown in FIG. 8, if the kickout counter is incremented **1630** and the counter threshold is not surpassed **1632**, the system moves on to evaluating **1634** how far the current rate is outside the acceptable range of rates. For example, the system may determine the difference between the current steering rate and the max or min of the acceptable range. When the difference in rates exceeds a particular threshold, the steering system reverts control **1650** to the operator. On the other hand, automatic steering control is maintained if the difference in rates does not exceed the difference threshold. In such cases operational control flows back **1660** to running the vehicle trajectory controller **1602** and awaiting a new kickout analysis **1610**.

According to various implementations, the threshold for evaluating the steering rate difference may be fixed or adjustable. In some cases the threshold has a default steering value that corresponds to 0.6 revolutions of the steering wheel per second.

As previously discussed, an operator may in some implementations determine the conditions or criteria for a kickout of automatic steering control. For example, an operator may select the value at which the kickout occurs based on a desired sensitivity for the kickout analysis. In some cases the kickout sensitivity is time based. For example, in some implementations a kickout system is based on the number of reported or current steering rates falling outside the expected range. In some cases the steering sensor reports its current steering rate at a fixed interval, thus making the kickout methodology time based. Accordingly, requiring more current steering rates to fall outside the expected range before kicking out decreases the kickout sensitivity by increasing the time required to kickout. Conversely, increasing the kickout sensitivity decreases the time to kick out.

FIG. 9 is a perspective view of another automatic steering system **100**, mounted within a vehicle according to an embodiment. As depicted in this particular example, the automatic steering system **100** includes a display unit **114** and an assisted steering device **104** mounted to the steering wheel **106** of the vehicle. A communications link **108** connects the assisted steering device **104** to the display unit **114** and a controller housed within. It should be appreciated that the controller may be located in one of many places according to various implementations. For example, in some cases the controller is a discrete physical component mounted separately from other components. In some cases the controller is mounted within and/or is integral to another component, such as the display unit **114**, the assisted steering device **104**, or another component.

According to various implementations, the controller (not shown in FIG. 9) is configured to determine a desired trajectory and control the assisted steering device **104** to steer the vehicle along the desired trajectory. As an example, in some cases the controller is configured to determine a vehicle curvature that will guide the vehicle from its current position to a new position along the desired trajectory. In various implementations the controller determines the vehicle curvature as a desired steering angle or steering wheel position. A controller is configured to then determine a turning rate (e.g., in terms of speed and direction) to bring the steering wheel from its current position to the desired position within a particular amount of time. It is appreciated that in various implementations, one or more controller is provided.

According to various implementations, the assisted steering device **104** includes a motor drive unit (MDU) with a drive motor configured to rotate the steering wheel in response to commands from the controller. The MDU also

13

includes a motor encoder that senses rotation of the motor using, for example, a Hall-effect sensor. The MDU generates and sends a feedback signal to the controller based on the sensor output. In various implementations the feedback signal includes the current encoder position and the current encoder rate. According to such implementations, the current encoder position corresponds to the current position of the motor and steering wheel and the current encoder rate corresponds to the speed and direction of the motor/steering wheel. In various implementations the controller also sends steering commands to the MDU in the form of desired encoder positions and/or rates. For example, in various cases the controller sends a desired encoder rate to the MDU to bring the encoder and the steering wheel from their current positions to new desired positions corresponding to a desired vehicle curvature.

In some cases, the current encoder rate is determined as a number of encoder counts per second. According to various implementations, the encoder has approximately 850 counts for one revolution of the steering wheel. In some such cases, the controller directs the MDU to adjust steering at up to a maximum rate of ± 1.5 revolutions (i.e., 1275 counts) per second, with the sign indicating whether to turn clockwise or counterclockwise. Further implementations are of course possible, regardless of limitations on max turning rate and how many counts there are per revolution.

In some implementations, the MDU may include additional features, characteristics, and capabilities known in the art. In various implementations the assisted steering device **104** is similar to or the same as one or more of the steering devices disclosed in U.S. application Ser. No. 16/921,828, entitled "Assisted Steering Apparatus and Associated Systems and Methods," filed Jul. 6, 2020, the entire content of which is hereby incorporated herein by reference. Those skilled in the art will further understand various methods and techniques for determining a desired trajectory and corrective steering angle. In some cases the controller may implement one or more techniques as taught in U.S. App. No. and in U.S. 63/026,208, entitled "Apparatus, Systems and Methods for Automatic Steering Guidance and Visualization of Guidance Paths," filed May 18, 2020, the entire content of which is hereby incorporated herein by reference.

FIG. **10** is a system diagram providing further detail about one example of an automatic steering system **200** according to various implementations. Like the system **100** depicted in FIG. **9**, the automatic steering system **200** includes a controller **202** in communication with an assisted steering device **104**. The assisted steering device **104** is configured to be mounted to a vehicle steering wheel (e.g., as shown in FIG. **9**) and to move the wheel in response to commands received from the controller **202**.

In the depicted example, the controller **202** is operationally integrated with a display unit **114**. One example of a suitable display unit is the display unit sold under the name InCommand® by Ag Leader Technology of Ames, Iowa. It is appreciated that certain of these displays **114** may include touchscreens, while others are equipped with interface components such as a keyboard, mouse, and/or other device for interacting with the system **200**.

According to various implementations, the controller **202** includes one or more processing and computing components for implementing various technologies disclosed herein and elsewhere. As shown in FIG. **10**, some of the components of the controller **202** can include a CPU or processor **210**, data storage **212**, an operating system **214**, and a graphical user interface (GUI) **216** accessible through, e.g., the display unit **114**. In this example the controller **202** also includes an

14

optional inertial measurement unit (IMU) **218**. It should be appreciated that the various system components are in operational communication with one another via wired or wireless connections and are configured to perform processes and execute commands to implement operations and features described herein.

In various implementations, the automatic steering system **200** is also operationally integrated with a GNSS or GPS unit **220**, such as a GPS **7500**, such that the system **200** is configured to input positional data for use in defining boundaries, locating the vehicle, plotting guidance, and the like, as will be readily appreciated. As an example, in some cases the controller **202** is configured to determine (e.g., estimate) the current position and/or trajectory of the vehicle using data from the GNSS/GPS unit **220** and/or the IMU **218**.

In various implementations the system **200** also includes a communications component **230** configured for sending and receiving data to one or more cloud-based locations **240** for processing and storage. As examples, data may be transmitted to and from a remote server **242**, database **244**, and/or other cloud computing components readily understood in the art. Such connections by the communications component **230** can be made wirelessly via understood internet and/or cellular technologies such as Bluetooth, Wi-Fi, LTE, 3G, 4 G, or 5 G connections and the like. It is understood that in certain implementations, the communications component **230** and/or cloud **240** components comprise encryption or other data privacy components such as hardware, software, and/or firmware security aspects. In various implementations, the operator or enterprise manager or other third parties are able to receive notifications such as adjustment prompts and confirmation screens on their mobile devices, and in certain implementations can review the plotted guidance paths and make adjustments via their mobile phones.

FIG. **11** is a flow diagram illustrating operation **300** of an automatic steering system according to various implementations. Among other things, the automatic steering system's operation includes running **302** a vehicle trajectory controller to generate a desired encoder position for adjusting the steering of a vehicle. The desired encoder position may correspond to, for example, a steering wheel position that will result in a particular steering curvature. As shown in FIG. **11**, the steering system also determines **304** motor information. In some cases, a motor drive unit of the system measures various properties and reports the information. In various implementations the determined motor information includes current encoder information. In some cases the current encoder information includes a current encoder position, a current encoder rate, or both. The desired encoder position is then used along with one or both of the current encoder rate and the current encoder position to run **306** a motor controller, which operates a steering motor to change the vehicle's steering trajectory. As an example, in some cases the desired encoder position is used along with one or both of the current encoder position and current encoder rate to determine a desired encoder rate for moving the encoder (and steering wheel) to the desired position.

In the depicted implementation, the operation of the automatic steering system includes a kickout analysis or process **310** that determines whether to disengage automatic steering control of the vehicle. In general, the kickout process **310** includes analyzing **320** a current encoder rate and desired encoder rate information. In various implementations, the desired encoder rate information is based on the desired encoder position and one or both of the current

15

encoder position and current encoder rate. As an example, in some cases the desired encoder rate information includes a desired encoder rate generated to run **306** the motor controller as previously discussed. In some cases the desired encoder rate information may be based on additional processing of a desired encoder rate and may include, for example, a history of desired encoder rates.

When the analysis determines that condition(s) for manual steering control are met, the automatic steering is disengaged and control reverts **350** to the operator. In some circumstances the analysis **320** may instead determine that manual steering criteria are not met. In these cases the operational flow loops back **360** to continue automatic steering control by running **302** the vehicle trajectory controller until a subsequent kickout routine **310** begins. It will be appreciated that in various implementations the analytical logic may be switched while retaining the same substantive analysis. For example, in some cases the analysis may instead ask if condition(s) for automatic steering control are met, with a true answer leading to the continuation of automatic steering control and a false answer leading to manual steering control.

According to various implementations, the automatic steering operation **300** is carried out by an automatic steering system such as one of the systems **100**, **200** shown in FIGS. **9** and **10**, respectively. In such cases the steering system controller(s) operate(s) as one or both of the motor controller and trajectory controller in FIG. **11**. In various implementations, one or more controllers are provided to accomplish the functionality described herein, as would be appreciated.

In some cases the steering system controller also determines **320** if manual steering conditions or criteria are met and reverts control **350** to the operator or loops control back **360** to the vehicle trajectory controller process **302**. According to various implementations, one or more additional controllers may carry out portions of the operation **300**. As just one example, in some cases an entirely separate vehicle trajectory controller determines the vehicle's position and generates a desired encoder position for correcting the vehicle's trajectory.

Various criteria, conditions, and/or analyses are used to evaluate the current encoder rate and the desired encoder rate information (e.g., desired encoder rate) in various implementations. As an example, in some cases the steering system evaluates the desired encoder rate information and the current encoder rate to determine how the vehicle's steering is tracking with the trajectory determined by the vehicle trajectory controller. In some cases differences between the desired and current encoder rates indicate that an operator is holding the steering wheel and/or overriding the automatic steering by physically turning the steering wheel. In such cases the system disengages the automatic steering control and reverts control **350** to the operator.

In some cases a lag exists between determining a desired encoder position and desired encoder rate and the steering motor responding accordingly. FIG. **12** is a graph **400** depicting plots of a desired or target encoder rate **402** and a current or actual encoder rate **404** according to various embodiments. As can be seen, the shape of the actual encoder rate **404** curve generally conforms to the target encoder rate **402** curve, but is shifted or delayed in time by some amount. Various implementations of an automatic steering system address such a lag by comparing a current encoder rate with desired encoder rate information that includes a history of desired encoder rates. Such an analysis

16

enables the steering system to determine if the encoder rates reported from the motor are tracking in the expected direction.

FIG. **13** depicts one example of comparing the current encoder rate with desired encoder rate information that includes a history of desired encoder rates according to various implementations. FIG. **13** depicts part of a kickout process **510** that generates a desired or acceptable range of target encoder rates and then determines **520** whether the current encoder rate is within the range of desired encoder rates. A determination that the current encoder rate is not inside (i.e., outside) the acceptable range is an indication that manual steering control may be desired and the system reverts steering control **550** to the operator. In contrast, a determination that the current encoder rate is within the acceptable range indicates that automatic steering should continue.

An acceptable range of desired or targeted encoder rates can be generated in various suitable manners according to the implementations disclosed and contemplated by this disclosure. In the implementation depicted in FIG. **13**, the automatic steering system updates **512** a rolling buffer with new desired encoder rates as they are received from the motor controller. In various cases the buffer contains the last twenty desired encoder rates, which corresponds to values over the last 0.4 seconds when the controller is run at 50 Hz. The buffer can collect a larger or smaller number of rates in various implementations. In some cases an expected range of encoder rates is then determined **514** using the maximum and minimum rates in the buffer.

In some implementations the range of encoder rates is expanded **516** by some amount to provide tolerance for system noise. As an example, the minimum rate of the range may be lowered by some amount and/or the maximum rate of the range may be increased by some amount. According to various implementations, the maximum rate is increased by 10% of its value and the minimum rate is lowered by 10% of its value. Different amounts of tolerance can be used depending upon the requirements or preferences of a particular implementation. In some cases a minimum tolerance of at least 50 encoder counts per second is used. As an example, a range minimum of -10 counts per second would result in a tolerance-adjusted minimum of -60 counts per second due to the 50 count per second minimum difference. Further implementations are of course possible, regardless of limitations on controller rate, length of the rolling buffer, minimum tolerance or otherwise.

Once the steering system has generated an expected range of acceptable encoder rates, the system determines **520** whether the current encoder rate is within the acceptable range and either maintains automatic control or reverts to manual control as noted above.

FIG. **14** is a flow diagram illustrating operation **600** of another automatic steering system according to various implementations. As with other examples described herein, the automatic steering system includes running **602** a vehicle trajectory controller to generate a desired encoder position for adjusting the steering of the vehicle. The steering system also determines **604** motor information to determine the current encoder position. The desired encoder position and one or both of the current encoder position and current encoder rate are then used to run **606** a motor controller, which operates a steering motor to change the vehicle's steering trajectory. In various implementations one or both of the current encoder position and current encoder rate and the desired encoder position are used to determine a desired encoder rate for adjusting the vehicle's steering. In some

cases running the vehicle trajectory controller, determining motor information, and running the motor controller are carried out in the same manner as the system operation 300 described with respect to FIG. 11.

The steering system operation also includes a kickout process or procedure 610 that determines whether to give manual control 650 to an operator or to loop back 660 to continue automatic steering control by running 602 the vehicle trajectory controller. As with the example of FIG. 13, the kickout process 610 includes generating an acceptable range of target encoder rates and then determining 620 whether the current encoder rate is within the acceptable range. To generate the acceptable range of encoder rates, the automatic steering system determines 614 the maximum and minimum rates from a rolling buffer 612 of desired encoder rates and then expands 616 the range by some amount to provide tolerance for system noise.

As discussed elsewhere, determining that the current encoder rate is outside the desired encoder range may indicate that control should revert to manual steering. In some cases, it may be desirable delay the switch to manual control in order to, for example, confirm the determination to disengage automatic control. As an example, in some implementations steering control is switched to a manual mode only after detecting multiple subsequent instances of the current encoder rate being outside the acceptable encoder range. The example kickout process 610 in FIG. 14 includes incrementing 630 a kickout counter each time the current encoder rate is outside the acceptable range of encoder rates. Upon exceeding a counter threshold 632, the steering system reverts control 650 to the operator.

According to various implementations, the counter threshold 632 can be set by the operator running the system. In some cases setting the counter threshold to a low value allows for easily and quickly switching to manual steering control. In some implementations a higher value for the counter threshold results in a longer kickout time but may result in fewer false kickouts. In some cases false kickouts occur when the steering wheel is bumped or an operator rests their hand on the wheel. In some cases field conditions bump the vehicle wheels. The movement can at times feed back into the steering column and be falsely interpreted by the system as the operator moving the steering wheel.

In some cases the counter threshold 632 can be fixed to require any number of kickouts suitable for a particular operating environment. In implementations such as those discussed above, an operator can set the counter threshold 632 to a value within a predefined range. In some cases the counter threshold can be any suitable value from a single instance to 50 or 100 or more instances. It should be appreciated as well that the desirable number for the threshold may depend upon the operating frequency of the system. In some cases the threshold can be set to a number from 3 to 43, which corresponds to a time range of 0.06 seconds up to 0.8 seconds when operating at 50 Hz.

As already noted, in some cases implementations of an automatic steering system will maintain automatic steering control, loop back to running the vehicle trajectory controller, and then conduct a new kickout analysis with updated encoder rates. For example, FIG. 14 illustrates how the steering system operation loops back 660 to running the vehicle trajectory controller 602 in certain circumstances. The kickout process 610 illustrates two situations in which automatic control is maintained in this way. In some cases the kickout process may determine 620 that the current encoder rate is actually within the acceptable range of desired encoder rates. Under these circumstances, the auto-

matic steering system sets the kickout counter to zero 622 and operational control flows back 660 to running the vehicle trajectory controller 602.

As another example, in some implementations a kickout process 610 further considers the degree to which a current encoder rate is outside the acceptable range of rates. In the example shown in FIG. 14, if the kickout counter is incremented 630 and the counter threshold is not surpassed 632, the system moves on to evaluating 634 how far the current rate is outside the acceptable range of rates. For example, the system may determine the difference between the current encoder rate and the max or min of the acceptable range. When the difference in rates exceeds a particular threshold, the steering system reverts control 650 to the operator. On the other hand, automatic steering control is maintained if the difference in rates does not exceed the difference threshold. In such cases operational control flows back 660 to running the vehicle trajectory controller 602 and awaiting a new kickout analysis 610.

According to various implementations, the threshold for evaluating the encoder rate difference may be fixed or adjustable. In some cases the threshold has a default value of 500 encoder counts per second, which corresponds to 0.6 revolutions of the steering wheel per second.

As previously discussed, an operator may in some implementations determine the conditions or criteria for a kickout of automatic steering control. For example, an operator may select the value at which the kickout occurs based on a desired sensitivity for the kickout analysis. In some cases the kickout sensitivity is time based. For example, in some implementations a kickout system is based on the number of reported or current encoder rates falling outside the expected range. In some cases the motor reports its current encoder rate at a fixed interval, thus making the kickout methodology time based. Accordingly, requiring more current encoder rates to fall outside the expected range before kicking out decreases the kickout sensitivity by increasing the time required to kickout. Conversely, increasing the kickout sensitivity decreases the time to kick out.

FIGS. 15-18 are graphs illustrating targeted and actual encoder rates in various circumstances according to some implementations. As previously discussed, in some implementations a determination that a current encoder rate is outside the acceptable range is an indication that manual steering control may be desired. In some cases the steering system may revert steering control to the operator as a result. In contrast, a determination that the current encoder rate is within the acceptable range indicates that automatic steering should continue.

FIG. 15 is a graph 700 depicting a plot of a target encoder rate 702 and an acceptable range of encoder rates 704. In various implementations, the acceptable range of encoder rates 704 corresponds to a specific target encoder rate, and thus the acceptable range 704 will shift along with the target rate. Also shown is a plot of a current, or actual, encoder rate 706 reported by a steering system motor encoder. According to various implementations, the extent of the acceptable range of encoder rates 704 is determined based on one or more preset criteria. For example, in some cases the acceptable range 704 has a length of time defined by a rolling buffer such as described with respect to FIG. 13. In some cases the minimum and maximum rates for the range 704 are centered about the value of the target rate by a preset amount. In some cases the minimum and maximum rates include a tolerance adjustment as previously discussed.

As seen in the graph 700, the actual encoder rate 706 stays relatively close to the target encoder rate 702 over time,

19

including within the acceptable encoder rate range **704**. In some implementations this is expected performance of the steering system and thus automatic control would be maintained. It can further be seen that the actual encoder rate **706** rises above the acceptable range around time **708**. According to some implementations, the deviation from the acceptable range would increment a kickout counter as described with respect to FIG. **14**. Exceeding a threshold count of rate deviations causes steering control to switch to the operator in various cases.

Turning to FIGS. **16-18**, plots of target and actual encoder rates are illustrated in graphs **710**, **720**, and **730**, respectively. As can be seen, each graph illustrates a case in which the actual encoder rate deviates from the target encoder rate and the acceptable rate range **704**. In FIG. **16** the actual encoder rate **716** remains constant for a time, which in various cases indicates that the operator has stopped the steering wheel. In FIGS. **17** and **18**, the target rates **726**, **736** deviate below and above, respectively, the acceptable range **704**, indicating that the operator has turned the steering wheel in various implementations.

Although the disclosure has been described with reference to preferred embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosed apparatus, systems and methods.

What is claimed is:

1. An automatic steering system for a vehicle, comprising: an assisted steering device configured to mount to a steering wheel of a vehicle, the assisted steering device comprising a motor and an encoder, wherein the motor is offset from a steering axis of the vehicle's steering wheel when mounted, the assisted steering device further configured to determine current encoder information; and a controller configured to operate the assisted steering device, the operation comprising: receiving the current encoder information from the assisted steering device; determining a desired encoder rate based on the current encoder information and a desired encoder position; operating the motor using the desired encoder rate; comparing a current encoder rate to desired encoder rate information; and disengaging automatic steering control when a result of the comparison meets a manual steering condition.
2. The automatic steering system of claim 1, wherein the desired encoder rate information comprises an acceptable encoder rate range.
3. The automatic steering system of claim 2, wherein the acceptable encoder rate range is based on a history of the desired encoder rate.
4. The automatic steering system of claim 2, wherein the manual steering condition comprises the current encoder rate being outside the acceptable encoder rate range.
5. The automatic steering system of claim 4, wherein the manual steering condition further comprises the current encoder rate being outside the acceptable encoder rate range for a number of consecutive comparisons exceeding a counter threshold.
6. The automatic steering system of claim 4, wherein the manual steering condition further comprises the current encoder rate being outside the acceptable encoder rate range by a difference exceeding a difference threshold.
7. The automatic steering system of claim 1, wherein the current encoder information comprises a current encoder position.

20

8. The automatic steering system of claim 1, wherein the current encoder information comprises the current encoder rate.

9. An automatic steering system for an agricultural vehicle, comprising:

- a steering sensor configured to mount to an agricultural vehicle with an assisted steering device and provide a feedback signal comprising vehicle steering information for automatically controlling the assisted steering device; and
- a controller operably coupled to the steering sensor and configured to:
 - receive the feedback signal and current steering information from the steering sensor;
 - determine desired steering rate information based on the current steering information and a desired steering position;
 - compare a current steering rate to the desired steering rate information; and
 - disengage automatic steering control based on comparing the current steering rate to the desired steering rate information;

wherein the assisted steering device comprises a motor and an encoder and is configured to mount to a steering wheel of the agricultural vehicle such that the motor is offset from a steering axis of the steering wheel.

10. The automatic steering system of claim 9, wherein the controller is further configured to update a counter based on comparing the current steering rate to the desired steering rate information and disengage the automatic steering control when the counter exceeds a counter threshold.

11. The automatic steering system of claim 9, wherein the controller is further configured to determine a difference between the current steering rate and the desired steering rate information and disengage the automatic steering control when the difference exceeds a difference threshold.

12. The automatic steering system of claim 9, wherein the desired steering rate information comprises a desired steering rate for operating the steering device.

13. The automatic steering system of claim 9, wherein the desired steering rate information comprises an acceptable steering rate range.

14. The automatic steering system of claim 13, wherein the controller is further configured to determine the acceptable steering rate range based on a history of a desired steering rate for operating the steering device.

15. The automatic steering system of claim 9, wherein the controller is operably coupled to the assisted steering device,

the steering sensor comprises the encoder, the current steering information comprises current encoder information,

the desired steering rate information comprises desired encoder rate information, and

the current steering rate comprises a current encoder rate.

16. A method for steering an agricultural vehicle, comprising:

- steering the agricultural vehicle with an assisted steering device comprising a motor and an encoder, the assisted steering device being mounted to a steering wheel of the vehicle with the motor offset from a steering axis of the steering wheel;
- receiving a feedback signal comprising current steering information with a controller configured to operate the assisted steering device;
- comparing the current steering information to desired steering information; and

21

disengaging automatic steering control when a result of the comparison meets a manual steering condition.

17. The method of claim **16**,
further comprising receiving the feedback signal from the
assisted steering device; 5
wherein the current steering information comprises a
current encoder rate; and
wherein the desired steering information comprises an
acceptable encoder rate range.

18. The method of claim **17**, wherein the manual steering 10
condition comprises the current encoder rate being outside
the acceptable encoder rate range.

19. The method of claim **18**, wherein the manual steering
condition further comprises the current encoder rate being
outside the acceptable encoder rate range for a number of 15
consecutive comparisons exceeding a counter threshold.

20. The method of claim **18**, wherein the manual steering
condition further comprises the current encoder rate being
outside the acceptable encoder rate range by a difference
exceeding a difference threshold. 20

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22